A Paper Design for a Grid-Forming UPS for AI Training Campuses, Sized from Certified Bounds

Sehaj Randhir Singh

Abstract: **This is a paper design. No hardware was built or tested, and nothing here is a measurement. Two earlier papers proved bounds for a battery-backed UPS between an AI training campus and the grid: grid power ramps no faster than a limit, the storage stays inside a computed power rating and energy window, and a headroom band is left for grid services. This paper turns those bounds into a bill of parts. For a 100 MW campus with a 20 MW/min ramp limit and a load floor that includes checkpoint dips, the certified requirement is 67.14 MW and a 2.796 MWh window. I size a shunt-connected SiC converter (1272 half-bridge modules in 31 blocks), a battery of 423,637 power-cell LFP cells (3.50 MWh nameplate), and a supercapacitor bank fed through a first-order split filter. I prove two facts about the split: the supercapacitor energy swing is at most $2\tau_s P_b$ and the battery ramp is at most $2P_b/\tau_s$. The supercapacitor pays off on battery throughput life, by a factor of about 16 on an assumed duty cycle, and does not pay off on efficiency: its own loss (411 kW) exceeds the battery loss it removes (375 kW). The sourced cost estimate is $14.5M, about $145 per kW of campus peak, with the unsourced items listed. The shunt design does not ride through voltage sags; a series variant costs more and is not certified.**

Index Terms: uninterruptible power supply, large loads, grid-forming converter, supercapacitor, silicon carbide.

I. Introduction

Synchronized AI training jobs swing a campus between a floor and a peak in about a second, and utilities and the North American Electric Reliability Corporation (NERC) now describe this as a reliability issue [4], [6]. Paper 1 proved guarantees for one battery between such a campus and the grid [1]; Paper 2 showed the guarantees survive a grid-service command inside a headroom band and studied fleets [2]. Neither says what to build. Here I do that on paper: topology, storage split, control stack, protection and cost.

Rules I follow. Every number is one of three kinds: (i) a saved result from Papers 1 and 2 or from `research/p3-blueprint/design_calc.py` (output in `results/design_calc.json`), (ii) a number read from a cited datasheet or report, or (iii) an assumption, marked *assumed*. If a choice does not pay off, I say so. The sections are the requirements (Sec. II), topology (III), the storage split (IV), the control stack (V), protection (VI), cost (VII) and limits (VIII).

II. Requirements From the Certificates

Paper 1 certifies a campus with load in $[P_{\rm lo}, P_{\rm hi}]$ and a grid ramp limit R. Sourced inputs [3]: $R = 20$ MW/min (the only numeric cap among surveyed utilities [7]), one-way efficiency $\eta = \sqrt{0.86} = 0.927$ [8], a 1.9 p.u./s fastest load ramp for about 250 ms on a 50 MW block, and a checkpoint dip to about 0.2 p.u. against a 0.6 p.u. training trough [4]. The Paper 1 branch-and-bound certificate, re-run with these values (`research/p0-sourcing`), gives the three design cases of Table I.

TABLE I
Certified requirements used for sizing. K is the reference design.

case	basis	P_b (MW)	window (MWh)
S	60 MW floor (P1 re-run)	40.28	1.006
K	checkpoint-dip floor	67.14	2.796
G	S + 15 MW headroom (P2)	75.28	4.638

I take K as the reference because the checkpoint dips are in the sourced data; a design for case S would be undersized for them. Cases K and G were certified separately, and I did not certify them together; a unit that has both a checkpoint-aware floor and grid-service headroom needs a new Paper 1 run, so G is an option and not a recommendation. Halving the ramp limit to 10 MW/min doubles the window (5.591 MWh for K) and does not change P_b, so R buys energy and not power.

III. Power Stage Topology

A. Shunt, not series

The certificate needs the storage to carry only the difference between load and the smoothed grid draw, at most P_b, not the whole load. A shunt (line-interactive) converter on the campus bus therefore needs P_b of converter, 67 MW for K, where a series (double-conversion) path would need the full 100 MW. The price is that a shunt converter cannot hold the load through a voltage sag (Sec. VI). Fig. 1 shows the single-line diagram.

B. Converter sizing

The device is the Wolfspeed CAB450M12XM3, a 1200 V, 450 A SiC half-bridge with $R_{\rm DS(on)} = 4.7\,\mathrm{m\Omega}$ at 175 °C and $E_{\rm on} = 24.4$ mJ, $E_{\rm off} \approx 8.35$ mJ at 600 V, 450 A [11]. Assumed: 800 V DC link, 480 V line-line output, 8 kHz switching, current derated to 0.67×450 A peak (213 A rms), and reactive capability $Q = 0.5P_b$. Then $S = \sqrt{67.14^2 + 33.57^2} = 75.1$ MVA, 90.3 kA rms at 480 V, 424 modules per phase, **1272 modules** in 31 blocks of 2.5 MVA. Per module, conduction loss is $I_{\rm rms}^2 R_{\rm DS} = 214$ W and switching loss $f_{sw} E_{\rm sw}(2/\pi)(I_{\rm pk}/450) = 149$ W, using $E_{\rm sw}$ scaled linearly in voltage to 800 V. Total 461 kW, 0.69 % of P_b at full power.

S. R. Singh is with Queens High School for the Sciences, New York, NY, USA.

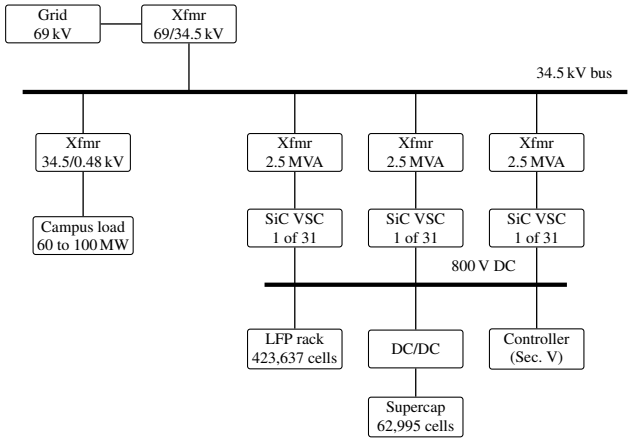


Fig. 1. Single-line diagram of the reference design (case K). The converter is split into 31 blocks of 2.5 MVA, each with its own transformer, SiC converter and share of the DC bus, battery and supercapacitor. Three blocks are drawn.

TABLE II
CONVERTER SIZING PER CASE (ASSUMED 480 V, 8 KHZ, $Q = 0.5P_b$; CAB450M12XM3 AT 213 A RMS).

case	MVA	modules	blocks	loss (kW)	% of P_b
S	45.0	765	19	277	0.69
K	75.1	1272	31	461	0.69
G	84.2	1425	34	517	0.69

Switching energy at 800 V is scaled, not read, so the switching figure is an estimate. The 480 V choice is a weakness: 90 kA at low voltage is why the design has 31 transformers; a medium-voltage cascaded converter would need different devices and I did not size one.

IV. BATTERY AND SUPERCAPACITOR SPLIT

A. Battery

The cell is the A123 ANR26650M1-B: 2.5 Ah at 3.3 V (8.25 Wh), 50 A continuous, 120 A for 10 s, 6 mΩ [9]. It is a power cell chosen because it is the only sourced datasheet in hand, not because it is the right cell for a grid site; a modern large-format LFP cell is the real answer and would change the cell count and cost. Cells needed for power are $P_b/(50\,\mathrm{A}\cdot 3.3\,\mathrm{V}) = 406{,}910$; cells needed for energy, with an assumed 80 % usable window, are $2.796\,\mathrm{MWh}/(0.8\cdot 8.25\,\mathrm{Wh}) = 423{,}637$. Energy binds for K, so the pack is **423,637 cells, 3.50 MWh nameplate, 32 t**. For S, power binds (244,122 cells, 2.01 MWh). For G, energy binds again (702,728 cells, 5.80 MWh). At $R = 10$ MW/min the K pack doubles to 847,122 cells and 6.99 MWh ($9.94M against $8.52M).

B. The split filter and what it provably does

Let p_b be the total storage power the certificate requires, $|p_b| \le P_b$. Send the battery p_{bat} from

$$\tau_s \dot{p}_{\mathrm{bat}} = p_b - p_{\mathrm{bat}}, \qquad p_{\mathrm{sc}} = p_b - p_{\mathrm{bat}}. \tag{1}$$

Claim. (a) $|p_{\mathrm{bat}}| \le P_b$; (b) $|\dot{p}_{\mathrm{bat}}| \le 2P_b/\tau_s$; (c) the supercapacitor energy $e_{\mathrm{sc}}(t) = \int_0^t p_{\mathrm{sc}}$ satisfies $e_{\mathrm{sc}}(t) = \tau_s\,(p_{\mathrm{bat}}(t) - p_{\mathrm{bat}}(0))$, so its range is at most $2\tau_s P_b$. *Proof.* The filter has impulse response $e^{-t/\tau_s}/\tau_s$ with unit L^1 norm, giving (a) with $p_{\mathrm{bat}}(0) = 0$. (b) follows from the equation since $|p_b - p_{\mathrm{bat}}| \le 2P_b$. (c) integrates $p_{\mathrm{sc}} = \tau_s \dot{p}_{\mathrm{bat}}$. □

Two consequences. The sum $p_{\mathrm{bat}} + p_{\mathrm{sc}} = p_b$ is unchanged, so the Paper 1 and 2 grid-side certificates are untouched as long as both devices can deliver what they are asked, and the energy loop must act on the *combined* stored energy. Second, the supercapacitor window the proof guarantees is large: for $\tau_s = 5$ s and case K it is $2\cdot 5\cdot 67.14\,\mathrm{MW} = 186.5\,\mathrm{kWh}$, which is 81,866 cells ($10.2M with its DC/DC). That is the price of sizing the supercapacitor to the proof. If the supercapacitor saturates, the battery simply sees more of the swing; the grid-side certificate is not harmed, only battery life.

TABLE III
BATTERY SIZING FOR THE THREE CERTIFIED CASES (A123 CELL, 8.25 WH, 50 A; 80 % USABLE WINDOW).

case	cells (power)	cells (energy)	binds	MWh	M$
S	244,122	152,425	power	2.01	5.08
K	406,910	423,637	energy	3.50	8.52
G	456,267	702,728	energy	5.80	10.32

TABLE IV
EFFECT OF THE SUPERCAPACITOR SPLIT, CASE K, $\tau_s = 5$ S, 1 S SWING, ASSUMED 50 % TRAINING DUTY. SUPERCAP LOSS IS FOR THE 4 S PERIOD.

	no SC	with SC
battery rms current per cell (A)	12.1	0.38
battery I^2R loss (kW)	375	0.4
supercap ESR loss (kW)	0	411
battery full cycles per yr	9,900	302
battery life (yr, cap 16)	0.97	16
oversized battery for 16 yr	57.7 MWh, $30.6M	n/a

C. Sizing to the sourced profile

I simulated the sourced swing (60 to 100 MW, 0.21 s edges from the 1.9 p.u./s ramp, periods 0.33, 1 and 4 s, matching the 0.2 to 3 Hz band in [6]); grid draw is held at the load mean, which is what the 90 s Paper 1 filter does to a fast square wave. With the supercapacitor off, the battery sees a 20 MW square wave with a 190 MW/s edge. With $\tau_s = 5$ s the worst of the three periods gives a supercapacitor peak of 25.8 MW, an energy swing of 10.4 kWh (the 4 s period), and a battery ramp of 5.2 MW/s (bound: 26.9). The supercapacitor cell is the Maxwell BCAP3000 P270: 3000 F, 2.7 V, 0.29 mΩ ESR, 0.55 kg, 5400 W/kg [10]; 3.04 Wh per cell, 2.28 Wh usable from 2.7 V down to 1.35 V. Cells required: 5,706 for energy (with an assumed 25 % margin), 8,671 for peak power, and **62,995 for rms current**, using the 150 A continuous rating and an assumed 0.75 of rated voltage. Current, not energy, sets the supercapacitor bank, so it is 35 t and an energy-oversized device.

D. Does it pay?

On life, yes, with a caution. PNNL gives 192,000 cycles at 5 % average depth of discharge [8], which I convert to 9,600

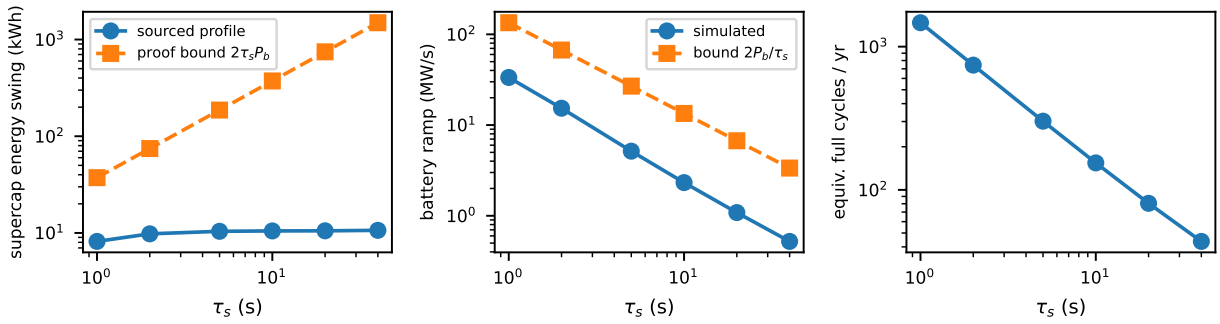


Fig. 2. Choice of τ_s for case K (`split_sweep.py`). Left: the sourced-profile supercapacitor swing flattens near 10.5 kWh once $\tau_s \geq 5$ s, while the proof bound keeps growing. Middle: battery ramp against the proved bound. Right: battery full cycles per yr at 1 s swing and 50 % duty. The 16-year throughput cap (9,600 cycles) is met from $\tau_s = 5$ s up; $\tau_s = 2$ s gives 12.9 years and $\tau_s = 1$ s gives 6.5.

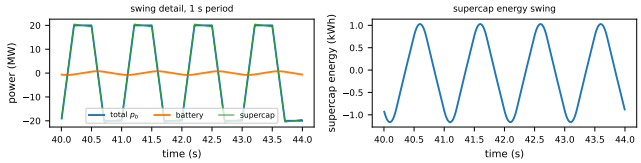


Fig. 3. Case K, $\tau_s = 5$ s, 1 s swing (`trace_fig.py`). Left: total storage power, battery share and supercapacitor share. The battery sees a few MW of ripple instead of a 20 MW square wave. Right: supercapacitor energy swing.

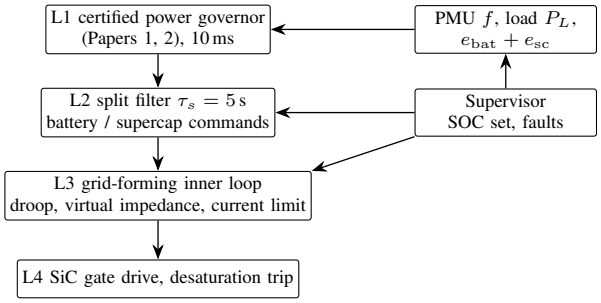


Fig. 4. Control layers. L1 is the only layer with a proof; the rest are conventional or checked by simulation only.

equivalent full cycles and use as a throughput cap, since the 1 s swings are only about 0.1 % depth and the table does not reach them. This is a crude proxy, not a rainflow model, and real shallow-cycle life may be longer or shorter. On that proxy the swing consumes the battery in about one year without the split, and the alternative of buying 57.7 MWh of battery to survive costs $30.6M against the $6.0M supercapacitor bank. On efficiency, no. The supercapacitor bank loses 411 kW to ESR (0.51 % of the 80 MW mean load) and removes 375 kW of battery loss, so the plant is 36 kW worse. The case for the split is battery life alone, and it depends on the duty-cycle assumption.

E. Why $\tau_s = 5$ s

Fig. 2 is the argument. A shorter τ_s gives the supercapacitor less energy to hold, but the battery then sees more of every swing: at $\tau_s = 1$ s its ramp is 33.5 MW/s and its throughput life under the proxy is 6.5 years. A longer τ_s does not shrink the sourced-profile supercapacitor swing (it saturates near 10.5 kWh) but it raises the energy window the proof guarantees, so a bank sized to the proof grows linearly with τ_s. So $\tau_s = 5$ s is the smallest value in the sweep that reaches the 16-year cap. This choice depends on the profile I assumed; a real campus with slower swings would push the supercapacitor energy requirement up, as the 4 s period already does against the 1 s one (10.4 kWh against 2.2 kWh).

F. Loss budget

At full certified power, the converter loses 461 kW (0.69 % of P_b, Sec. III-B). In steady training, supercapacitor ESR loss is 411 kW and battery I^2R is 0.4 kW. Against the 80 MW mean load the converter and supercapacitor together are about 0.5 % to 1 % depending on how often the converter works at full power; I did not model transformer, cooling and auxiliary losses. The one-way efficiency in Papers 1 and 2 (0.927, from the PNNL 86 % AC-AC round trip) is far lower than these figures because it includes battery electrochemistry; I kept it for the certificates and did not tighten it.

V. Control Stack Around the Certificates

Fig. 4 shows the layers; the cycle times are assumptions.

L1 is the Paper 1 law (rate-clipped first-order low-pass of the load plus a saturated loop that restores the *settled* state of charge), with the Paper 2 frequency term limited to the headroom band. Its proofs are in continuous time with an exact power command. A discrete-time implementation at a 10 ms cycle is not covered by those proofs, and I did not prove a sampled version. **L2** is the split filter of Sec. IV-B. **L3** is a voltage-source inner loop with droop and virtual impedance, as in the standard grid-forming literature [15]; it must track the L1 power command with the converter lag Paper 2 assumed (50 ms, assumed, not measured). Whether a real grid-forming loop holds that lag on a weak feeder is the largest unproved link in this design. **L4** is the gate-drive protection of Sec. VI. A **supervisor** drops to a conservative limiter (rate clip only, no frequency term) if the combined energy leaves the certified set or the PMU signal is lost; a plain limiter drains the same

battery (Paper 1, 60 of 140 profiles), so this is a safe mode of limited duration, not a substitute.

A. Operating modes

Four modes follow from the supervisor. *Normal*: L1 tracks the load with the certified ramp limit, L2 splits the command. *Grid service*: the Paper 2 frequency term is active inside the headroom band (case G only). *Degraded*: the supervisor has lost the PMU signal or the energy estimate left the certified set, so only the rate clip runs; the battery can drain on adversarial bursts, so this mode needs an alarm and a time limit. *Island or sag*: the shunt converter cannot hold the load and the facility's own protection decides; this is the gap Sec. VI discusses.

B. Why the numbers look the way they do

Three results are not obvious from the bounds alone. First, the battery is sized by two nearly equal constraints. With these cells the power requirement gives 406,910 cells and the energy window gives 423,637, so the 20 MW/min ramp limit sits close to the point where the pack stops being energy-limited. At the Paper 1 value of 10 MW/min the energy term would dominate and the pack would double. Second, the cost is mostly in the per-kW items of the PNNL breakdown, because the certified energy window is only 2.8 MWh for a 67 MW requirement, a duration of about 2.5 minutes, far below the 2 h column I borrowed. Third, the supercapacitor bank is set by the rms current of a 1 s to 4 s square wave, not its energy, because the cell current rating of 150 A limits how much a 2.7 V cell can deliver continuously. A cell with a higher current rating would cut the bank size and cost, and I did not look for one.

VI. Protection

Converter. SiC modules carry integrated temperature sensing and need desaturation detection at the gate driver; per-block AC breakers and a DC contactor with fuses isolate a failed block. With 31 blocks of 2.5 MVA the installed real power at $Q = 0.5P_b$ is $31 \cdot 2.5/1.118 = 69.3$ MW. One lost block leaves 67.08 MW against the 67.14 MW requirement, a shortfall of 0.06 MW, so the design is N-1 only by a hair and a spare block (32 blocks, about 3 % more converter) is the honest choice. I did not simulate the certificate under a reduced P_b.

Storage. Cell and string management for 423,637 cells, thermal-runaway propagation testing to UL 9540A and siting to NFPA 855 [16], [17]. Supercapacitor strings need voltage balancing per cell. Fusing and arc-flash studies are required and not done.

Grid events. On 10 July 2024 about 1,500 MW of data center load moved to backup after six faults in 82 s, each 42 to 66 ms, because a common UPS scheme transfers after three disturbances within a minute [5]. The supervisor must not count disturbances inside the ride-through envelope; I used the NERC PRC-029-1 voltage and frequency ride-through tables and IEEE 2800 as the reference envelope [13], [14], though a UPS is not a generator and these standards do not formally apply to it (assumption). A shunt converter cannot fix the root problem: during a sag the load's own supplies see the depressed voltage. Holding the load requires a series path at the full 100 MW. The energy for the six faults is small, 11.0 kWh at 100 MW, but the power is not: 33,671 supercapacitor cells at 5400 W/kg ($2.19M), and a 100 MW converter at the PNNL $73.05/kW costs $7.31M against $4.90M for the shunt one, plus continuous double-conversion loss. This option is not certified by Papers 1 and 2, and I do not recommend it before a series-path certificate exists.

TABLE V
Cost, reference case K, $\tau_s = 5$ s (USD).

item	M$	basis
battery, energy items	1.43	408.35 $/kWh × 3495 kWh
battery, power items (incl. PCS)	7.09	105.61 $/kW × 67.14 MW
supercap cells	4.09	62,995 × $65 (assumed)
supercap DC/DC	1.88	73.05 $/kW × 25.75 MW
total	14.49	$145/kW of 100 MW peak

VII. Cost Estimate

Battery cost uses the PNNL 2021 LFP column for 10 MW, 2 h [8]. Per-kWh items (DC block 176.21, DC BOS 43.40, systems integration 51.62, EPC 62.33, project development 74.79) total $408.35/kWh. Per-kW items (power conversion 73.05, controls 7.75, grid integration 24.81) total $105.61/kW. Supercapacitor cost is the cell price (assumed $65, from a DigiKey single-piece list near 59 EUR [12]) plus a DC/DC at the PNNL power-conversion rate.

Sensitivities, all computed: doubling the DC block and BOS rate for power-cell chemistry (assumed) gives $15.3M; doubling the supercapacitor cell price gives $18.6M; $R =$ 10 MW/min gives $15.9M; case S with its battery is $5.08M plus the supercapacitor bank. The PNNL figures are 2021 dollars for a grid system, not for 20C power cells. **Not costed:** the SiC premium over PNNL's power-conversion rate, the 31 low-voltage transformers (PNNL's grid integration item may cover part), a cooling plant for 420,000 cells, buildings and the N+1 block. The total is therefore a floor.

VIII. What Is Calculated, What Is Assumed, and Where It Is Weak

Calculated or proved: the requirements (Papers 1 and 2), the split-filter bounds (Sec. IV-B), cell, module and cost counts from the datasheet and PNNL numbers, the profile simulation, and loss totals. **Assumed:** 800 V and 480 V levels, 8 kHz, 0.67 current derating, $Q = 0.5P_b$, 80 % usable battery, 0.75 supercapacitor current voltage factor, $65 per supercapacitor cell, 50 % training duty, $\tau_s = 5$ s, the 1 s to 4 s swing shape, the 50 ms converter lag and 10 ms control cycle.

Weaknesses. (1) The certificates are continuous-time and ideal-tracking; no part of the grid-forming loop, the filter implementation or the sampled controller is proved. (2) The battery life gain rests on a throughput proxy and a duty assumption, and the supercapacitor does not improve efficiency. (3) The supercapacitor bank is set by cell current rating; a

TABLE VI
EVERY INPUT OF THE DESIGN, WITH ITS ORIGIN. S: SOURCED FROM A CITED DOCUMENT; C: CALCULATED HERE OR IN PAPERS 1 AND 2; A: ASSUMED.

quantity	value	type	origin
grid ramp limit R	20 MW/min	S	Southern Company limit in [7]
fastest load ramp	1.9 p.u./s, 250 ms	S	NERC white paper [4]
checkpoint dip, trough	0.2, 0.6 p.u.	S	NERC white paper, read off a plot (approximate)
swing frequency band	0.2 to 3 Hz	S	Choukse et al. [6]
one-way efficiency η	0.927	S	$\sqrt{0.86}$, PNNL AC-AC round trip [8]
required P_b, window (K)	67.14 MW, 2.796 MWh	C	Paper 1 certificate re-run, checkpoint floor
required P_b, window (G)	75.28 MW, 4.638 MWh	C	Paper 2 unit certificate, headroom 15 MW
LFP cell data	8.25 Wh, 50 A, 6 mΩ	S	A123 datasheet [9]
supercap cell data	3000 F, 0.29 mΩ, 0.55 kg	S	Maxwell datasheet [10]
SiC module data	450 A, 4.7 mΩ, 24.4 mJ	S	Wolfspeed datasheet [11]
cost per kWh and kW	\$408.35, \$105.61	S	PNNL Fig. 4.3, 10 MW 2 h LFP, 2021 [8]
fault sequence	6 faults, 42 to 66 ms, 82 s	S	NERC incident review [5]
split-filter bounds	$2\tau_s P_b$, $2P_b/\tau_s$	C	proof in Sec. IV-B
cell counts, module counts, losses	Tables III, II	C	`design_calc.py`
supercap cell price	\$65	A	DigiKey single-piece list near 59 EUR [12]; bulk price unknown
usable battery window	80 %	A	typical planning value, not sourced
DC link, AC voltage, f_{sw}	800 V, 480 V, 8 kHz	A	design choice
current derating	0.67	A	design choice
reactive capability	$Q = 0.5P_b$	A	design choice
τ_s	5 s	A	chosen from the sweep of Fig. 2
training duty	50 %	A	unsourced
swing waveform	trapezoid, 60 to 100 MW	A	simplified from NERC and Choukse
converter lag, control cycle	50 ms, 10 ms	A	Paper 2 assumption; not measured
supercap margin and voltage factor	25 %, 0.75	A	design choice

TABLE VII
DESIGN DECISIONS AND WHETHER THEY PAID OFF.

choice	paid off?	evidence
shunt converter, not series	yes, with a cost	67 MW instead of 100 MW of converter; cannot ride through sags
supercap split	life only	battery life 0.97 to 16 yr on a proxy; efficiency 36 kW worse
$\tau_s = 5$ s	yes	smallest value reaching the cap (Fig. 2)
tighter ramp limit	cheap	$R = 10$ adds \$1.4M, energy only
grid-service headroom (G)	costly	75.3 MW, 5.80 MWh, \$10.3M against \$8.5M, not certified with K
series ride-through	not shown	\$7.31M converter against \$4.90M, uncertified

different cell would change cost and size by a large factor. (4) The cell is a 2.5 Ah 26650 power cell, which is unrealistic for a 3.5 MWh pack. (5) The shunt design cannot ride through faults. (6) Cases K and G were not certified together. (7) The simulation uses a symmetric trapezoid wave, not a recorded trace, and grid draw is idealized as the load mean. (8) The cost is a floor, from 2021 grid-storage numbers, not a quote.

IX. WHAT WOULD BE TESTED NEXT

A paper design earns trust only through tests, and I list them in the order I would run them. (1) A controller hardware-in-the-loop run of L1 to L4 with the 50 ms converter lag replaced by a real converter model, to see whether the certified ramp bound survives a sampled, saturating implementation. (2) A rainflow study of battery state of charge on a recorded training trace, replacing the throughput proxy of Sec. IV-D; this decides whether the supercapacitor split is worth about \$6.0M. (3) A single-module bench test of the CAB450M12XM3 at 800 V to replace the scaled switching energy. (4) A fault study of one lost block against the 0.06 MW shortfall. (5) A series-path certificate, which would turn the ride-through option from a cost line into a design. None of these is done.

Reproducibility. The calculations are in `research/p3-blueprint/design_calc.py` and `split_sweep.py`; outputs are `results/design_calc.json` and `results/split_sweep.json`. They run in seconds on a laptop, so nothing needed Kaggle.

X. CONCLUSION

The certified bounds fix a converter of about 67 MW and a storage window of 2.8 MWh. The power requirement, not the energy requirement, dominates cost: \$7.09M of the battery's \$8.52M is per-kW. The ramp limit buys energy cheaply (\$1.4M for halving R). The supercapacitor split is justified only by battery life, on a stated proxy. The next step that matters is a real certificate for the grid-forming inner loop, and a measurement.

REFERENCES

[1] S. R. Singh, "Certified grid-side behavior of an uninterruptible power supply for AI training loads," 2026, repository `ai-ups-cert`, `paper/main.tex`.

[2] S. R. Singh, "Fleets of certified AI-load UPS units for oscillation damping and frequency support on the Kundur two-area system," 2026, repository `ai-ups-cert`.

[3] Repository `ai-ups-cert`, `research/p0-sourcing/SOURCING.md`, 2026.

[4] NERC Large Loads Task Force, "Characteristics and risks of emerging large loads," white paper, Jul. 2025.

[5] NERC, "Incident review: Considering simultaneous voltage-sensitive load reductions," Jan. 2025.

[6] E. Choukse *et al.*, “Power stabilization for AI training datacenters,” arXiv:2508.14318, 2025.
[7] R. Zahedi, A. Zamani, and R. Anilkumar, “Best practices for large load interconnections: A North American perspective on data centers,” arXiv:2601.12686, 2026.
[8] V. Viswanathan *et al.*, “2022 grid energy storage technology cost and performance assessment,” PNNL-33283, Pacific Northwest National Laboratory, Aug. 2022.
[9] A123 Systems, “Nanophosphate high power lithium ion cell ANR26650M1-B,” datasheet.
[10] Maxwell Technologies, “MC power series 650 F to 3,000 F ultracapacitors, BCAP3000 P270,” datasheet, doc. 1009361.10.
[11] Wolfspeed, “CAB450M12XM3 1200 V, 450 A silicon carbide half-bridge module,” datasheet.
[12] DigiKey, product listing for BCAP3000 P270 K05 (Maxwell Technologies), accessed Oct. 2026; single-piece price near 59 EUR on the Irish site.
[13] NERC, “PRC-029-1, Frequency and voltage ride-through requirements for inverter-based resources,” reliability standard.
[14] IEEE Std 2800-2022, “Interconnection and interoperability of inverter-based resources interconnecting with associated transmission electric power systems.”
[15] J. Rocabert, A. Luna, F. Blaabjerg, and P. Rodríguez, “Control of power converters in AC microgrids,” *IEEE Trans. Power Electron.*, vol. 27, no. 11, pp. 4734–4749, 2012.
[16] UL LLC, “UL 9540A, Test method for evaluating thermal runaway fire propagation in battery energy storage systems.”
[17] NFPA, “NFPA 855, Standard for the installation of stationary energy storage systems.”